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Grade	**12.50** out of 100.00

Question 1

Incorrect

Mark 0.00 out of 30.00

Create a Java class called `Product` that represents an inventory item. The class must follow these requirements:

✅ Properties:

- Private `String productName`
- Private `String category`
- Private `double price`

✅ Constructor:

- A constructor that takes `String productName`, `String category`, and `double price`.

✅ Methods:

- `public String getProductCode()`
  Returns a code: First 3 letters of category (UPPERCASE) + Last 3 characters of name (UPPERCASE).
  Example: "Electronics", "Laptop" → "ELETOP"
- `public String getFormattedPrice()`
  Returns the price formatted as `"$XX.XX"` using two decimal places.
- `public String getSummary()`
  Returns: "Code: <CODE>, Item: <NAME IN TITLE CASE>, Price: <FORMATTED PRICE>"

🔤 Title Case Note:

Example: "gaminG laptOp" → "Gaming Laptop"

Hint-1: Use `s.substring(s.length() - 3)` for the last 3 characters.

Hint-2: Use `String.format("$%.2f", price)` for price formatting.

For example:

Test	Result

`Product p = new Product("laptOp Pro", "Electronics", 1200.5);`
`System.out.println(p.getProductCode());`
`System.out.println(p.getSummary());`
	`ELEPRO`
	`Code: ELEPRO, Item: Laptop Pro, Price: $1200.50`

`Product p = new Product("gaMing mOuse", "Accessories", 25.0);` `System.out.println(p.getSummary());`	`Code: ACCUSE, Item: Gaming Mouse, Price: $25.00`

Answer: (penalty regime: 0 %)

Reset answer

```
1 public class Product {
2
3     public string getProductCode(){
4
5
6     }
7 }
```

Syntax Error(s)

```
__tester__.java:3: error: cannot find symbol
    public string getProductCode(){
           ^
  symbol:   class string
  location: class Product
__tester__.java:17: error: constructor Product in class Product cannot be applied to given types;
        Product p = new Product("laptOp Pro", "Electronics", 1200.5);
                    ^
  required: no arguments
  found:    String,String,double
  reason: actual and formal argument lists differ in length
__tester__.java:19: error: cannot find symbol
System.out.println(p.getSummary());;
                   ^
  symbol:   method getSummary()
  location: variable p of type Product
3 errors
```

Show/hide question author's solution (Java)

```java
public class Product {
    private String productName;
    private String category;
    private double price;

    public Product(String productName, String category, double price) {
        this.productName = productName.trim().replaceAll("\\s+", " ");
        this.category = category.trim();
        this.price = price;
    }

    public String getProductCode() {
        String catPart = category.substring(0, 3).toUpperCase();
        String namePart = productName.substring(productName.length() - 3).toUpperCase();
        return catPart + namePart;
    }

    public String getFormattedPrice() {
        return String.format("$%.2f", price);
    }

    public String getSummary() {
        String titleCaseName = "";
        for (String word : productName.split(" ")) {
            if (!word.isEmpty()) {
                titleCaseName += word.substring(0, 1).toUpperCase() + word.substring(1).toLowerCase() + " ";
            }
        }
        return "Code: " + getProductCode() + ", Item: " + titleCaseName.trim() + ", Price: " + getFormattedPrice();
    }
}
```

Incorrect

Marks for this submission: 0.00/30.00.

Question 2

Partially correct

Mark 2.50 out of 5.00

Which of the following statements are correct? Please select all that apply.

Select one or more:

- A. A reference variable is an object.
- B. A reference variable references to an object.
- C. A data field in a class must be of a primitive type.
- D. A data field in a class can be of an object type. ✔ A is wrong because a reference variable is not an object, it is a reference that points to an object. C is incorrect because a class may have a data field of an object type such as String.

A is wrong because a reference variable is not an object, it is a reference that points to an object. C is incorrect because a class may have a data field of an object type such as String.

The correct answers are: A reference variable references to an object., A data field in a class can be of an object type.

Question 3

Incorrect

Mark 0.00 out of 5.00

What is the value of times displayed?

```
public class Test {
 public static void main(String[] args) {
 Count myCount = new Count();
 int times = 0;

 for (int i = 0; i < 100; i++)
 increment(myCount, times);

 System.out.println(
 "myCount.count = " + myCount.count);
 System.out.println("times = "+ times);
 }

 public static void increment(Count c, int times) {
 c.count++;
 times++;
 }
}

class Count {
 int count;

 Count(int c) {
 count = c;
 }

 Count() {
 count = 1;
 }
}
```

Select one:

- A. 101
- B. 100
- C. 99 ✖
- D. 98
- E. 0

See Listing 9.10.

The correct answer is: 0

Question 4

Incorrect

Mark 0.00 out of 5.00

Analyze the following code:

```
public class Test {
 public static void main(String[] args) {
 double radius;
 final double PI = 3.15169;
 double area = radius * radius * PI;
 System.out.println("Area is " + area);
 }
}
```

Select one:

- A. The program has a compile error because a constant PI is defined inside a method.
- B. The program has compile errors because the variable radius is not initialized.
- C. The program compiles and runs fine.
- D. The program has no compile errors but will get a runtime error because radius is not initialized. ✖

radius is a local variable. It is used in radius * radius * PI, but not initialized.

The correct answer is: The program has compile errors because the variable radius is not initialized.

Question 5

Incorrect

Mark 0.00 out of 5.00

Analyze the following code.

```
public class Test {
 int x;

 public Test(String t) {
 System.out.println("Test");
 }

 public static void main(String[] args) {
 Test test = new Test();
 System.out.println(test.x);
 }
}
```

Select one:

- A. The program has a compile error because Test does not have a default constructor.
- B. The program has a compile error because you cannot create an object from the class that defines the object.
- C. The program has a compile error because System.out.println method cannot be invoked from the constructor. ✖
- D. The program has a compile error because x has not been initialized.

Note that a default no-arg constructor is provided only if no constructors are explicitly defined in the class. In this case, a constructor Test(String t) is already defined. So, there is no default no-arg constructor in the Test class.

The correct answer is: The program has a compile error because Test does not have a default constructor.

Question 6

Incorrect

Mark 0.00 out of 5.00

You should add the static keyword in the place of ? in Line _____ in the following code:

```
1  public class Test{
2      private int age;
3
4      public ? int square (int n){
5          return n*n;
6      }
7
8      public ? int getAge(){
9          return age;
10     }
11 }
```

Select one:

- A. in line 4
- B. in line 8
- C. in both line 4 and line 8 ✖
- D. none

The square method should be static because it does not reference any instance data or invoke any instance method.

The correct answer is: in line 4

Question 7

Correct

Mark 5.00 out of 5.00

Analyze the following code:

```
public class Test {
 public static void main(String[] args) {
 double[] x = {2.5, 3, 4};
 for (double value: x)
 System.out.print(value + " ");
 }
}
```

Select one:

- A. The program displays 2.5, 3, 4
- B. The program displays 2.5 3 4
- C. The program displays 2.5 3.0 4.0 ✔ See the first paragraph.
- D. The program displays 2.5, 3.0 4.0
- E. The program has a syntax error because value is undefined.

See the first paragraph.

See the first paragraph.

The correct answer is: The program displays 2.5 3.0 4.0

Question 8

Correct

Mark 5.00 out of 5.00

Analyze the following code.

```
class TempClass {
  int i;
  public void TempClass(int j) {
    int i = j;
  }
}

public class C {
  public static void main(String[] args) {
    TempClass temp = new TempClass(2);
  }
}
```

Select one:

- A. The program has a compile error because TempClass does not have a constructor with an int argument. ✔ The program would be fine if the void keyword is removed from public void TempClass(int j).
- B. The program compiles fine, but it does not run because class C is not public.
- C. The program has a compile error because TempClass does not have a default constructor.
- D. The program compiles and runs fine.

The program would be fine if the void keyword is removed from public void TempClass(int j).

The program would be fine if the void keyword is removed from public void TempClass(int j).

The correct answer is: The program has a compile error because TempClass does not have a constructor with an int argument.

Question 9

Incorrect

Mark 0.00 out of 30.00

Write a Java program to simulate a digital wallet transfer service that converts 5 different amounts from **USD to EUR**.

The program should:

- Read **5** USD amounts from the user.
- Ask the user for the service type: "**Standard**" or "**Express**".
- Use a method `processTransfers(double[] amounts, String type)` to calculate the payout.

Business Logic:

- **Exchange Rate:** 1 USD = 0.92 EUR.
- **Fixed Service Fee:** Always deduct **$0.50** from the USD amount first.
- **Commission:**
  - "Standard": 1% commission on the converted EUR.
  - "Express": 3% commission on the converted EUR.
- **Validation:** If an amount is ≤ $0.50, print "Insufficient funds for fee."

Formula: `Result = ((USD - 0.50) * 0.92) * (1 - Commission Rate)`

IMPORTANT HINTS:

- **String Comparison:** In Java, do **NOT** use `==` to compare Strings. Use the `.equals()` method (e.g., `type.equals("Standard")`).
- **Rounding:** Use `Math.round(value * 100.0) / 100.0` for two decimal places.
- **Array Handling:** Your method should return a `double[]` array containing the results.
- **Invalid Service Type:** Handle "Invalid service type." for any other input.

For example:

Input	Result
Standard 100 0.40 50.50 10 200	Service Type: Standard Enter 5 amounts: 100.0 0.4 50.5 10.0 200.0 Source: $100.0 : Payout: 90.62 EUR Source: $0.4 : Insufficient funds for fee. Source: $50.5 : Payout: 45.54 EUR Source: $10.0 : Payout: 8.65 EUR Source: $200.0 : Payout: 181.7 EUR
InvalidType 10 20 30 40 50	Service Type: InvalidType Enter 5 amounts: 10.0 20.0 30.0 40.0 50.0 Invalid service type.
Express 80 0.49 90.50 100 20	Service Type: Express Enter 5 amounts: 80.0 0.49 90.5 100.0 20.0 Source: $80.0 : Payout: 70.95 EUR Source: $0.49 : Insufficient funds for fee. Source: $90.5 : Payout: 80.32 EUR Source: $100.0 : Payout: 88.79 EUR Source: $20.0 : Payout: 17.4 EUR

Answer: (penalty regime: 0 %)

```
2  public class Q9{
3
4      public static void main(String [] args){
5          Scanner inputs = new Scanner(System.in);
6
7          String serviceType = inputs.nextLine();
8          System.out.print("Service Type: " + serviceType);
9          System.out.println("");
10         System.out.print("Enter 5 amounts: ");
11
12         for(int i = 0; i<5; i++){
13           double[] amounts = inputs.nextDouble();
14           System.out.print(amounts[i] + " ");
15         }
16         System.out.println("");
17
18         processTransfer(amounts, serviceType);
19     }
20     //static????????
21     //public static processTransfer(double[] amounts, String type) ??
22     public static void processTransfer(double[] amounts, String type){
23         double result = 0.0;
```

```
        double [] amount;
        if(!(type.equals("Standard") || type.equals("Express"))){
            System.out.print("Invalid service type.");
        }
        else if(type.equals("Standart")){

            for(int j=0; j<5; j++){
                result[j] = ((amounts[j] - 0.50)*0.92)*(1 - 0.01);

                System.out.print("Souce: $" + amounts[j] + " : Payout: " + Math.round((results[j] * 100.0) / 100.0) + " EUR"
            }
            result = 0.0;
        }
        else if(type.equals("Express")){
            for(int k= 0; k<5; k++){
                result[k] = ((amount[k] -0.50) * 0.92) *(1 - 0.03);

                System.out.print("Source: $" + amount[k] + " : Payout: " + Math.round((results[k] * 100.0) / 100.0) + " EUR"
            }

        }

    //    return result;
    }



}
```

Syntax Error(s)

```
Q9.java:13: error: incompatible types: double cannot be converted to double[]
          double[] amounts = inputs.nextDouble();
                                                ^
Q9.java:18: error: cannot find symbol
        processTransfer(amounts, serviceType);
                        ^
  symbol:   variable amounts
  location: class Q9
Q9.java:31: error: array required, but double found
                result[j] = ((amounts[j] - 0.50)*0.92)*(1 - 0.01);
                     ^
Q9.java:33: error: cannot find symbol
                System.out.print("Souce: $" + amounts[j] + " : Payout: " + Math.round((results[j] * 100.0) / 100.0) + " EUR");
                                                                                    ^
  symbol:   variable results
  location: class Q9
Q9.java:39: error: array required, but double found
                result[k] = ((amount[k] -0.50) * 0.92) *(1 - 0.03);
                     ^
Q9.java:41: error: cannot find symbol
                System.out.print("Source: $" + amount[k] + " : Payout: " + Math.round((results[k] * 100.0) / 100.0) + " EUR");
                                                                                    ^
  symbol:   variable results
  location: class Q9
6 errors
```

```
import java.util.*;
public class Main {

   public static double [] processTransfers(double [] amounts, String type){
      double [] result=new double[5];
      double commission=0.0;

      if(type.equals("Standard"))
         commission =0.01;
      else if (type.equals("Express"))
         commission =0.03;
      else
         return result;
```

```
        for(int i=0; i<amounts.length;i++){
            if (amounts[i]<=0.5)
                result[i] = -1.0;
            else
                result[i] = ((amounts[i] - 0.50) * 0.92) * (1 - commission);
            result[i] = Math.round(result[i] * 100.0) / 100.0;
        }

        return result;
    }

    public static void main(String[] args){
        Scanner input = new Scanner(System.in);
        double [] amounts = new double[5];
        double [] result = new double[5];
        String serviceType = input.next();
        System.out.println("Service Type: " + serviceType);
        System.out.print("Enter 5 amounts: ");
        for(int i=0; i<amounts.length; i++){
            amounts[i]=input.nextDouble();
            System.out.print(amounts[i] + " ");
        }
        System.out.println();
        result = processTransfers(amounts, serviceType);
        if (result[0]==0)
            System.out.println("Invalid service type.");
        else
            for(int i=0; i<amounts.length; i++){
                if (result[i]<0)
                    System.out.println("Source: $"+amounts[i] + " : Insufficient funds for fee.");
                else
                    System.out.println("Source: $"+amounts[i] + " : Payout: " +result[i] + " EUR");
            }
    }
}
```

Show/hide question author's solution (Java)

```java
import java.util.Scanner;

public class WalletSystem {
    public static void processTransfers(double[] amounts, String type) {
        double commission;
        if (type.equals("Standard")) {
            commission = 0.01;
        } else if (type.equals("Express")) {
            commission = 0.03;
        } else {
            System.out.println("Invalid service type.");
            return;
        }

        for (double usd : amounts) {
            if (usd <= 0.50) {
                System.out.println("Source: $" + usd + " : Insufficient funds for fee.");
                continue;
            }
            double afterFixedFee = usd - 0.50;
            double eurBase = afterFixedFee * 0.92;
            double finalEur = eurBase * (1 - commission);
            double rounded = Math.round(finalEur * 100.0) / 100.0;
            System.out.println("Source: $" + usd + " : Payout: " + rounded + " EUR");
        }
    }

    public static void main(String[] args) {
```

```
        Scanner sc = new Scanner(System.in);
        double[] vals = new double[5];
        System.out.print("Service Type: ");
        String type = sc.nextLine();
        System.out.println(type);
        System.out.print("Enter 5 amounts: ");
        for (int i = 0; i < 5; i++) {
            vals[i] = sc.nextDouble();
            System.out.print(vals[i] + " ");
        }
        System.out.print("\n");
        processTransfers(vals, type);
    }
}
```

Incorrect

Marks for this submission: 0.00/30.00.

Question 10

Incorrect

Mark 0.00 out of 5.00

Analyze the following code:

```
public class Test {
 public static void main(String[] args) {
 System.out.println(max(1, 2));
 }

 public static double max(int num1, double num2) {
 System.out.println("max(int, double) is invoked");

 if (num1 > num2)
 return num1;
 else
 return num2;
 }

 public static double max(double num1, int num2) {
 System.out.println("max(double, int) is invoked");

 if (num1 > num2)
 return num1;
 else
 return num2;
 }
}
```

Select one:

- A. The program cannot compile because you cannot have the print statement in a non-void method.
- B. The program cannot compile because the compiler cannot determine which max method should be invoked.
- C. The program runs and prints 2 followed by "max(int, double)" is invoked.
- D. The program runs and prints "max(int, double) is invoked" followed by 2. ✖
- E. The program runs and prints 2 followed by "max(double, int)" is invoked.

This is known as ambiguous method invocation. See the last Note box in this section.

The correct answer is: The program cannot compile because the compiler cannot determine which max method should be invoked.